Residual basis reduction and smaller Diophantine consistency equations

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Abstract

This report studies explicit polynomial equations whose unsolvability is provably equivalent in ZFC to the syntactic consistency of ZFC. The main new method changes the basis of quadratic residual equations to reveal linear consequences, then eliminates their unit-coefficient coordinates. It reduces both the dimension and the coefficient cost of a direct proof-calculus construction. Further results combine a shift-free descending-factorial certificate with residual reduction, replace unnecessarily large Pell transport denominators, and remove redundant arithmetic gates from small-variable root certificates. All comparisons specify the domain and the measured cost. The unbounded arguments use explicitly retained ordinary arithmetic and refutation calibrations. Exact finite checks certify the supplied polynomials and transformations; they are not a new end-to-end proof-assistant certificate, a consistency proof, or a theorem of global optimality.

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1 The problem and the comparison boundary

For a fixed integer polynomial P and a stated domain D, the target is

$$\mathrm{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

Here consistency means absence of a finite syntactic refutation in the specified effective proof system. The statement is obtained by proving solvability equivalent to existence of such a refutation and then negating both sides. It does not decide whether either side holds.

There is no single total order of simplicity. We measure the number of quantified variables, their domain, total degree, number of quadratic residuals where applicable, support of the fully collected polynomial, maximum absolute integer coefficient, and live binary operations in a specified straight-line evaluator. Addition, subtraction and multiplication each cost one, including squares and multiplication by fixed constants. Reusing a register is free. Fixed numerical ports are not variables, but their entire coherent tuples must be specified.

The immutable repository comparison is `e0c98c629bf10d30b695884f6b1d2ab361bdf513` in ProveIt. The archive retains the source identities and distinguishes actively accepted repository constructions from newly submitted incoming packages. In particular, the incoming *Cofinal Anchors* and *Certificate Reductions* packages are stronger comparators than our preceding report. Our reductions of their literal equations are explicit; the word "incoming" does not imply an independent external review or repository adoption. Attribution and inherited mathematical hypotheses are preserved.

1.1 Selected results

Domain and variant	Variables	Degree	Rows	Terms	Height
$\mathbb{Z}$, dimension	278	4	188	12884	12227
$\mathbb{Z}$, balanced	279	4	189	5483	7200
$\mathbb{Z}$, sparse	279	4	189	5202	12662
$\mathbb{Z}$, fewer terms	284	4	194	4270	10864
$\mathbb{Z}$, sparsest listed	306	4	216	3870	14512
$\mathbb{N}$, balanced	229	4	179	5607	65892965
$\mathbb{N}$, smaller lower height	228	4	178	9522	560804902
$\mathbb{N}$, smaller sparse	228	4	178	7512	2445412949744

Here $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{Z}$ denotes unrestricted integers. The 279-variable balanced equation improves the incoming 282-variable comparison on every reported varying measure: its parent has 192 rows, 5491 terms and height 128446. The 278-variable equation goes one variable further at a support cost. The other integer comparisons are to the matched 287-variable/197-row/4313-term and 309-variable/219-row/3922-term incoming equations, both of height 128446. These are direct proof-calculus reductions, not absolute variable-count records across all coefficient encodings.

Domain and branch	Variables	Exact degree	Binary operations
Strictly positive, $D = XY$, squared weight	10	626	77
Strictly positive, $D = XY$, product weight	10	464	78
Strictly positive, $D = q^4$, product weight	10	460	79
Strictly positive, partial normalization	10	384	80
$\mathbb{Z}$, auxiliary root and odd guard	14	86	112
$\mathbb{Z}$, strong root and odd guard	14	94	110
$\mathbb{Z}$, exact recompiled predecessor	13	156	125

The new 14-variable degree-86 equation improves both degree and evaluator cost over the preceding degree-88/119-operation form. Degree 94 is a lower-cost companion, not a degree improvement. A degree-88/115-operation exact recompilation is also retained as a check. The 13-variable entry preserves the entire earlier polynomial and only shortens its evaluator. No smaller fixed-port payload or fully expanded coefficient height is claimed for these circuit branches.

2 Changing residuals without changing solutions

Lemma 2.1 (Elementary residual operations)**.** *Let $R = (R_1, \ldots, R_m)$ be integer polynomials. Replacing R_i by $R_i + kR_j$, for $i \neq j$ and $k \in \mathbb{Z}$, preserves their common zero set on every fixed coordinate domain. Its inverse replaces the new R_i by $R_i - kR_j$. Consequently, over $\mathbb{Z}^n$ or $\mathbb{N}^n$, the sums of the squares of the old and new residuals have precisely the same zeros.*

Proof. The forward and inverse substitutions are polynomial identities and leave all other rows unchanged. A sum of integer squares vanishes if and only if every summand vanishes. The assertion about sums of squares uses this ordered-domain fact; it is not asserted over the complex numbers. □

Lemma 2.2 (Exposing an affine coordinate)**.** *Suppose exact elementary row operations produce a row $\varepsilon x_j + L(\boldsymbol{x}_{\neq j})$, where $\varepsilon \in \{1, -1\}$ and L is affine with integer coefficients. Substitute $x_j = -\varepsilon L$*

in every row and delete the resulting zero row and the coordinate x_j. The remaining system has a bijective correspondence of integer zeros with the original system, with the displayed formula as inverse coordinate restoration. Quadratic residuals remain quadratic. The assertion for natural-number coordinates additionally requires a proof that the restored coordinate is nonnegative.

Proof. At every zero the displayed row uniquely specifies x_j. Conversely, substitution followed by restoration makes that row zero and makes every retained row equal to its substituted counterpart. Affine substitution does not increase degree. Repeat in the recorded order for several eliminations and invert in reverse order. □

These elementary facts motivate a different optimization problem from changing the encoded proof calculus or its Pell witnesses. Take the matrix of quadratic coefficients of the residual list. Its integer row combinations can have a vanishing quadratic part even when none of the individual rows is affine. Such combinations expose otherwise hidden coordinates. Afterwards a second residual-basis search can lower the support or coefficient height of the sum of squares. The final polynomial usually changes, while its zero set does not. Every accepted operation is recorded and replayed exactly; heuristic choices made during the search are not part of the proof.

3 Hidden affine consequences: quartics in 278 and 279 integers

The new direct-calculus contribution is a reduction of the incoming 282-coordinate quartic to 279 coordinates without a support or height penalty, and to 278 coordinates at a stated support tradeoff. The mechanism is not a new number-theoretic assumption: elementary changes of residual basis expose affine identities that the original gate presentation hides. The corresponding 287- and 309-coordinate parents also lose three coordinates. Every coordinate remains an unrestricted integer.

Equation	Variables	Rows	Monomials	Height
Incoming dimension parent	282	192	5491	128446
Incoming grouped parent	287	197	4313	128446
Incoming sparse parent	309	219	3922	128446
New dimension tradeoff	278	188	12884	12227
New balanced	279	189	5483	7200
New sparse 279	279	189	5202	12662
New sparse 284	284	194	4270	10864
New sparse 306	306	216	3870	14512

All eight displayed equations have exact degree four. The two 279-coordinate equations improve every varying metric of the matched 282-coordinate parent; the 284- and 306-coordinate equations do likewise for their matched parents. The 278-coordinate point sacrifices support for one fewer variable. These are literal finite endpoints, not global-minimality or literature-priority claims.

3.1 Equivalence lemmas

Residual basis. If $i \neq j$ and $k \in \mathbb{Z}$, replacing R_i by $R_i + kR_j$ leaves the common zero set unchanged: one implication is substitution, and the other follows by subtracting kR_j after using $R_j = 0$. The inverse uses $-k$, so each such operation is unimodular. The sum-of-squares polynomial generally changes; its zero set does not.

Integral coordinate chart. The substitution $x_t = z_t + L(z)$, with L integral affine and independent of z_t, is a bijection of integer lattices. Its inverse is $z_t = x_t - L(x)$. A sequence is inverted in reverse order. Quadratic degree is preserved.

Unit affine projection. If one residual is $\epsilon x_t + L = 0$, with $\epsilon \in \{1, -1\}$ and L integral affine and independent of x_t, substitute $x_t = -\epsilon L$ into every other residual and delete this row and coordinate. Restriction maps old zeros to new zeros; the displayed integral formula gives the unique inverse extension. Repeated extensions are evaluated in reverse projection order. Degree remains at most two.

Since $\sum_i R_i^2 = 0$ over $\mathbb{Z}$ exactly when every $R_i = 0$, these three facts prove the corresponding equivalence for a single quartic. No sign guard, divisibility condition or cofinal choice is introduced.

3.2 Credited preconditioning and the new identities

Six previously published-in-repository coordinate shears are transferred to the three incoming parents. This preconditioning is credited prior work, not the novelty claim. With zero-based coordinate indices, the successive old-coordinate substitutions are

$$\begin{aligned} &x_{21} \leftarrow x_{21} - 2x_{10} - 2x_{11}, && x_{23} \leftarrow x_{23} - 8x_{22}, \\ &x_{38} \leftarrow x_{38} - 8x_{37}, && x_{35} \leftarrow x_{35} - 4x_{10} - 5x_{12} - 2x_{25} - x_{27}, \\ &x_{27} \leftarrow x_{27} - x_{10} - x_{12}, && x_{25} \leftarrow x_{25} - x_{11} - x_{12}. \end{aligned}$$

They are executed in the displayed order, not simultaneously. The upstream certificates, exact source hashes and commit are retained. Each forward and inverse map is independently checked here; no semantic acceptance claim for the upstream V-R1 endpoint is needed.

Let Q_i denote the residuals after this preconditioning. Exact coefficient collection in each of the three parents gives

$$\begin{aligned} Q_{21} - Q_6 &= 4 + 4x_{10} + 8x_{11} + 4x_{21} - x_{22} + x_{37}, \\ Q_{73} - Q_{71} - 4Q_{70} - 8Q_{68} &= 12 - 6x_3 - 8x_{84} - 4x_{85} - x_{86} + x_{89}, \\ Q_{122} - Q_{120} &= 3 - 2x_{107} + 2x_{151} - x_{167} + x_{169}. \end{aligned}$$

These are polynomial identities, not consequences inferred from sample solutions. Replace rows 21, 73 and 122 by the left sides, using five elementary row shears. The donor rows are unchanged. The unit-affine projection lemma then gives

$$\begin{aligned} x_{37} &= x_{22} - 4 - 4x_{10} - 8x_{11} - 4x_{21}, \\ x_{89} &= x_{86} - 12 + 6x_3 + 8x_{84} + 4x_{85}, \\ x_{169} &= x_{167} - 3 + 2x_{107} - 2x_{151}. \end{aligned}$$

The deleted coordinates are named `add.145`, `mul.346` and `base4.T`. This proves the drops $(282, 192) \to (279, 189)$, $(287, 197) \to (284, 194)$ and $(309, 219) \to (306, 216)$ in coordinates and rows.

3.3 One further projection and the 278-coordinate point

The already affine row Q_{90} has unit coordinate x_{105}, named `add.380`. It is not counted as a newly discovered affine identity. Its unique restoration is

$$\begin{aligned} x_{105} = x_{10} + x_{11} + 4x_{21} + 2x_{24} + 4x_{26} + 2x_{29} + 2x_{31} + 4x_{32} \\ + 2x_{34} + 2x_{36} + 4x_{39} + 2x_{40} + 2x_{43} + 2x_{47} + 4x_{68} + 2x_{69}. \end{aligned}$$

Applying this additional projection to the dimension parent gives 278 coordinates and 188 rows. The dense replacement explains the larger collected support. No new semantic lemma is needed.

3.4 The fixed quartics and their logical meaning

Each endpoint is defined exactly by

$$P(\mathbf{z}) = \sum_{i=0}^{m-1} r_i(\mathbf{z})^2, \qquad \mathbf{z} \in \mathbb{Z}^n,$$

where its complete list of integral quadratic r_i is in `equation.json`; complete collected coefficients are in `expanded_polynomial.json`. The package also supplies every row as human-readable text and TeX, the full coordinate-order table, and a complete affine map restoring the original parent coordinates.

Following the projections, a frozen list of elementary residual row shears selects each stated support/height profile. The lists are fully explicit in the endpoint certificates. Their proof is the residual-basis lemma applied successively, not the search heuristic that found them. The deterministic builder runs no optimizer. Exact expansion confirms degree four and that every declared coordinate occurs.

Theorem. Each supplied new equation has an integer zero if and only if its matched incoming parent has an integer zero. Hence, composing with the parent's ordinary C222/ZF/ZFC refutation calibration, ZFC proves that integer unsolvability of each new equation is equivalent to Con(ZFC).

Proof. For a new zero, undo the final row shears to recover all projected residual equations. Restore the deleted coordinates in reverse order; the deleted equations vanish identically and all remaining equations hold. Undo the affine-exposure row shears, then evaluate the coordinate charts in reverse execution order. This produces an old zero. In the other direction, use the inverse coordinate bijection, the exact affine equalities, restriction to retained coordinates, and the forward row shears. Every step is integral and reversible on zeros. The final logical equivalence is exactly the inherited parent calibration. This argument does not prove consistency or claim a new end-to-end kernel-checked Lean theorem.

3.5 Verification and the limit of this particular method

An independent standard-library checker imports no producer. It checks coordinate inverses, elementary row inverses, unit coefficients, the identical zero of every deleted row, absence of deleted coordinates, the exact coordinate renumbering, every collected coefficient and metric, and coordinate liveness. It also reconstructs every original coordinate as an affine polynomial in the final ones and checks every original residual pullback as a polynomial identity. Five corrupted-artifact controls are rejected. Normal and `python -O` runs pass, including fresh byte-identical rebuilds. The printed residual text and TeX are parsed back to their complete coefficient lists.

For the 278-coordinate endpoint, the quadratic parts of the 188 residuals have rank 188 modulo 1000003. Thus they also have rank 188 over $\mathbb{Q}$: a nonzero modular minor cannot have zero integer determinant. No nonzero rational linear combination of this final residual basis is affine. The parent's corresponding rank is 188, and the four displayed independent affine combinations account for its four-dimensional kernel. This finite obstruction concerns this row span only, not nonlinear ideal consequences or global variable minimality. The optional rank replay is supplied separately.

4 Shift-free descending factorials and cofinal natural quartics

The new natural-domain branch combines the descending factorial identity from the incoming *Certificate Reductions* packet with the cofinal semantic anchor and strict amplification exponent from Round 17. Its new step removes the artificial positive shifts on both the factorial and the CRT-carrier coordinates, deriving their positivity in the correct order from the already present polynomial rows.

Endpoint	Variables	Rows	Terms	Height
Round 17 balanced	229	179	9184	83954634
New projected precursor	229	179	6596	83879378
New balanced, row-reduced	229	179	5607	65892965
Round 17 smaller	228	178	9866	2445412949744
New smaller, lower height	228	178	9522	560804902
New smaller, sparser	228	178	7512	2445412949744

Every row concerns a quartic over independently nonnegative integers. The two 228-variable variants use the identical new semantic graph and only change the certified elimination order; all restorations are proved nonnegative. In particular, no integer-only coordinate projection is being applied to natural variables.

4.1 The exact polynomial and logical statement

The balanced polynomial is

$$P_{229}(x_0,\dots,x_{228}) = \sum_{j=0}^{178} R_j(x_0,\dots,x_{228})^2, \qquad (x_0,\dots,x_{228}) \in \mathbb{N}^{229}.$$

The full literal residual list is supplied in `balanced229/residuals.tex` and `equation.json`; every collected coefficient is in its `expanded_polynomial.json`. The analogous files in `low_height228/` and `sparse228/` define their complete 178-row polynomials. A JSON term $[c,[i,j,\dots]]$ means $cx_ix_j\cdots$; an empty index list denotes a constant. These polynomials contain no binomial, factorial, floor, power, or Pell primitive.

With the retained ordinary numerical C222/ZF/ZFC refutation calibration, each polynomial has a natural zero exactly when a finite syntactic ZFC refutation exists. Consequently ZFC proves the equivalence between its no-natural-zero sentence and Con(ZFC). This neither proves consistency nor supplies a new end-to-end kernel proof or a global minimality theorem. The coefficient verification below is finite; the unbounded arithmetic steps are proved separately.

4.2 Semantic positivity replaces two coordinate shifts

Keep $t = t_0 + 12$, but now use unshifted natural coordinates F, n, B, r, s and the three polynomial rows

$$n + 1 = F^2, \qquad V = BF + r, \qquad B = r + s + 1. \tag{1}$$

The first exact power request is $V = n^t$. Its retained natural modulus coordinate T_n obeys

$$T_n + n^2 + 1 = 2An.$$

This literal equality excludes $n = 0$, even before any power or factorial is interpreted. Thus $F \geq 2$ and $n = F^2 - 1 \geq 3$. The new row $n + 1 = F^2$ is deliberately not marked as a defining gate with globally nonnegative restoration $F^2 - 1$.

The remaining exact requests, with the same self-sized radix as Round 17, are

$$V = n^t, \quad N = B^x, \quad q = (NS)^x, \quad D_t = q^t, \quad D_y = q^{y+1}, \quad D_x = q^x, \tag{2}$$

where

$$S = 1 + C + A_a + A_b + A_c + A_d + A_S + A_{S'} + A_u + A_v.$$

All representatives are positive. The shared frame is the sum of every distinct power base, exponent and output, together with every modular top index, plus 15. There are thirteen nonanchor seeds and eleven modular requests. The scale equations give $A \geq (W + 1)^W$, and the one-row anchor described below authenticates all indices before the powers are recovered. No factorial identity is needed for this stage: all atoms are natural and all intended exponents are positive, since $t \geq 12$, $x \geq F + 1 \geq 3$, and $y \geq F + 1 \geq 3$.

Lemma 4.1 (Shift-free bootstrap)**.** *Once the exact powers are recovered, the rows* (1) *imply*

$$B > F \geq 2, \qquad V < B(F + 1) \leq B^2 \leq N \leq q.$$

In particular, the first binomial can be decoded without assuming $F = t!$ or assuming that its output is positive in advance.

Proof. The remainder rows give $B \geq 1$ and $V < B(F + 1)$. For $F \geq 2$,

$$n = F^2 - 1 \geq F + 1, \qquad V = n^t \geq n^2 > F(F + 1).$$

Therefore $B > F$, and integrality gives $B \geq F + 1$. Since $x \geq 3$, $B^2 \leq B^x = N \leq (NS)^x = q$. For $0 \leq j \leq t$, $\binom{n}{j} \leq n^j \leq n^t = V < q$, with $\binom{n}{j} = 0$ when $j > n$. Thus every relevant coefficient is carry-free. □

The root-modulus divisibility needed at this early point is also already available. The raw endpoints are

$$y \geq F + 1, \qquad x \geq F + 1, \qquad x = 1 + (1 + (t + 1)y)u_t.$$

As $x \geq 3$, $u_t \geq 1$, giving $x \geq 2 + (t + 1)y > \max(t + 1, y + 1)$. Hence $M = Nq^x$ is divisible by qD_t and by all required range denominators multiplied by N, without yet knowing $F = t!$. The exact modular digit now gives $B = \binom{n}{t}$. Since $B \geq 1$, $n \geq t$, and the two remainder rows say $F = \lfloor n^t / \binom{n}{t} \rfloor$.

Lemma 4.2 (Descending fixed point)**.** *If $t \geq 3$, $F \geq 2$, and $n = F^2 - 1 \geq t$, then*

$$F = \left\lfloor \frac{n^t}{\binom{n}{t}} \right\rfloor \quad \Longleftrightarrow \quad F = t!.$$

Proof. Put $T = t!$, $U = t(t - 1)/2$, and

$$H = \frac{n^t}{\binom{n}{t}} = T \prod_{j=0}^{t-1} (1 - j/n)^{-1} \geq T.$$

If $F = \lfloor H \rfloor$, then $F \geq T$. The inequality $T - 1 > U$ holds at $t = 3$ and persists by induction: from $T \geq U + 2$ one gets $(t + 1)T - 1 \geq (t + 1)(U + 2) - 1 > U + t$. Consequently $n \geq T^2 - 1 = (T + 1)(T - 1) > (T + 1)U$. The elementary inequality $\prod(1 - a_j) \geq 1 - \sum a_j$ gives

$$T \leq H \leq \frac{Tn}{n - U} = T + \frac{TU}{n - U} < T + 1.$$

Thus $F = T$. Conversely the same estimates at $F = T$ prove that $\lfloor H \rfloor = T$. □

4.3 The descending carrier and its correctly oriented index

Lemma 4.3. *Let $F = t!$, $t \geq 3$. Then*

$$d_i = F^2/i - 1 \quad (1 \leq i \leq t)$$

are pairwise coprime positive integers greater than one, every prime divisor of each d_i exceeds t, and $\prod_i d_i = \binom{F^2-1}{t} = B$.

Proof. Since $i \mid F$, $d_i \equiv -1 \pmod{F}$, excluding all primes at most t. A common prime of d_i, d_j for $i \neq j$ would divide $id_i - jd_j = j - i$, impossible for a prime exceeding t. The product identity is $\prod_i (F^2 - i)/i = \binom{F^2-1}{t}$. □

The literal index equality is now

$$C + 1 = F^2 + Bq_C, \qquad \text{equivalently } C = n + Bq_C, \tag{3}$$

not the rising-carrier predecessor's differently signed row. Modulo d_i, it identifies $C = i - 1$. Its quotient is natural in completeness: $F \geq 6$, $3 \leq t \leq n - 2$, and $B = \binom{n}{t} \geq \binom{n}{2} > n$. Every positive representative congruent to $n \pmod{B}$ is at least n.

4.4 The shared Pell anchor and recovery chronology

For $A \geq 2$, define the ordinary Pell sequences by

$$(A + \sqrt{A^2 - 1})^j = X_j(A) + Y_j(A)\sqrt{A^2 - 1}.$$

The complete retained three-norm certificate indexes a single principal pair at the derived label J. The simultaneous row is

$$Y_J(A) = J + \sum_i (Y_i - k_i)^2 + (A - 1)Q, \qquad Q \in \mathbb{N}. \tag{4}$$

Here the notation $Y_J(A)$ is its proved meaning, not a primitive of the polynomial. Before indexing, the row supplies the anchor's omitted inequality $J \leq Y_*$. Its full private witnesses remain natural.

For clarity, the retained cap argument applies whenever $W \geq 3$, $2 \leq J < W$, $A - 1 > W^3$, at most $W - 2$ seeds occur, and $1 \leq k_i < W$. Pell classification writes $Y_i = Y_{j_i}(A)$. If $j_i \geq J$, then $Y_i \geq Y_J(A) = V_* \geq 2A > 2W$, so $(Y_i - k_i)^2 \geq (V_* - W + 1)^2 > V_*$, contradicting (4). Thus $j_i < J < W$. Reduction modulo $A - 1$, using $Y_j(A) \equiv j \pmod{A - 1}$, shows that

$$0 \leq \sum_i (j_i - k_i)^2 < W^3 < A - 1$$

is divisible by $A - 1$, hence zero. Every index is authenticated.

Conversely let $m = \max k_i$ and $z = Y_m(A)$. If $J \geq 2m$, then

$$Y_J(A) \geq 2X_m(A)Y_m(A) > 2(A - 1)z^2.$$

The sum of actual squared errors is at most rz^2; since $2(A - 1) - r > J$, the numerator defining Q is positive and divisible by $A - 1$. This yields its natural value.

The scale rows are

$$L^2 - ((W + 1)^2 - 1)(WZ)^2 = 1, \quad Z = z_0 + 1, \quad A = L + \sigma.$$

Their positive Pell index is divisible by W, so $A \geq X_W(W+1) \geq (W+1)^W$. Nonanchor seeds retain natural consecutive coordinates Y, Z' with $Y^2 + (Z')^2 - 2AYZ' = 1$. Each has an output incidence $bY = Z' + R$, $R \geq 0$, $1 \leq b < A$. Thus $Y > 0$, $Z' < AY$, and $X = AY - Z' > 0$ gives the ordinary principal norm. The anchor argument therefore applies to these ordinates too.

For every exact request the authenticated pair satisfies

$$X_k(A) + (b - A)Y_k(A) = m + (2Ab - b^2 - 1)h.$$

The Pell congruence gives $m \equiv b^k \pmod{2Ab - b^2 - 1}$. The frame has $b, k, m < W$, while its lower scale makes the modulus larger than $W^W + W$; hence $m = b^k$. Canonical positive Pell residues and their least nonnegative residues give nonnegative quotients. The common-root equation

$$2A(q+1) = (q+1)^2 + 1 + Mh_0, \qquad M = Nq^x,$$

gives the analogous congruence for all modular numerator requests. Lemma 4.1 then decodes the first binomial, and only then Lemma 4.2 supplies $F = t!$.

4.5 Local bounds, soundness, and constructive completeness

The endpoint equalities now imply $R = 2 + t + x + y + F < 2x$ and $x > 2^{32}$. The retained literal local majorant has $E = 2^{23}$. At $x = 2^{32}$, $(2x)^E < 2^x$; the ratio bound

$$(1 + 1/x)^E < \frac{1}{1 - E/x} < 2$$

propagates the inequality by induction. Thus each required local source value is $< R^E < 2^x$. The retained label majorant has $\ell = 2^{22}$, with $\ell(t+1) < F < x$ for every padded $t \geq 12$. These are unbounded estimates, not numerical search cutoffs. The strict beta quotient ranges $u, v < x$ are complete because their canonical denominators are at least two.

Every independent representative is $< S$, so its binomial coefficients through lower index x are $< S^x < q$. The derived label has $J \leq S^\ell$, making all its needed coefficients through $t+1$ also $< S^x$. All eleven modular requests are therefore carry-free. For each row $i - 1$, choose $p \mid d_i$ and $m = v_p(d_i)$. The retained one-direction range lemma

$$p^e \mid \binom{a}{k} \implies a \bmod p^e \in [0, k-1]$$

recovers the small local values modulo p^{xm}. The index row (3) distinguishes $i - 1$. The nonnegative local source value is $< 2^x \leq p^{xm}$, so its zero residue forces its actual value to vanish. The derived label is similarly bounded. The unchanged beta and Horn-flow arithmetic yields a finite C222 refutation, and the named retained calibration gives ZFC refutability.

For completeness start with a finite refutation, expand to its proof tree, and pad $t \geq 12$ beyond every local parameter and label. Use the inherited beta construction for x, y. Set $F = t!$, $n = F^2 - 1$, $B = \prod_i d_i$, $N = B^x$. Lemma 4.2 supplies the ordinary Euclidean remainder and complementary slack in (1). Choose the nine positive independent representatives modulo d_i^{2x}, initially in $(0, N^2]$, and increase only A_a by N^2. All local polynomial congruences to precision $2x$ are preserved. A falling-factorial numerator has a term of valuation at least $2xm$, while division by $k!$ loses at most $k - 1 \leq x - 1$; hence the required binomial is divisible by p^{xm}. This genuine precision argument supplies range completeness; no false converse of the range lemma is used.

The nominal indices $t, y+1, x, n$ are at most $x^2+x < 2^x \le N$; every independent representative is at most $2N^2$. The Cantor-derived label obeys

$$J \ge 2(A_a + N^2)^2 > 2N^4 > 4N^2 \ge 2 \max_i k_i.$$

Thus the cofinal shift supplies the anchor's completeness hypothesis without another polynomial row. Choose S, q, denominators and W, L in that order. Since $N \mid q$, $\gcd(q+1, M) = 1$; the linear common-root congruence has gcd one or two, and in the latter case its right side is even. Choose A arbitrarily large in that solution class. All scale and root quotients can be natural. Canonical Pell pairs, fresh private anchor witnesses, Q from (4), and Euclidean digit witnesses complete the graph. The choices are acyclic.

4.6 A reversible row-basis improvement

The final balanced endpoint further replaces the 179 precursor residuals by 42 explicit elementary shears. Each operation, recorded as (i, j, k) with $i \ne j$, is

$$R_i \longleftarrow R_i + kR_j, \qquad k \in \mathbb{Z} \setminus \{0\}.$$

Its inverse is the same operation with $-k$, applied in reverse order. Thus the old and new residual vectors vanish on exactly the same tuples. Their sums of squares need not have equal values, but have identical zero sets over $\mathbb{N}^{229}$. No coordinate is changed, projected, or made signed. The resulting quartic has 5,607 collected monomials and coefficient height 65,892,965. The precursor's 6,596 terms and height 83,879,378 are retained as a control in `projected229/`. The complete operations, target residuals and exact metrics are in `row_basis_certificate.json`; `check_row_basis.py` checks every forward and inverse row identity, every collected coefficient, coordinate liveness, and the refreshed embedded endpoint metrics. It rejects six mutations, including deliberately stale metric metadata.

4.7 Exact export, natural restoration, and reproducibility

The independent affine checker validates all 613 defining gates, 392 nonnegative affine restorations, all 285 exported source rows, and all 4,606 raw coefficients. The balanced precursor contraction has 106 eliminations; both 228-variable variants have 107. Each restoration is an integer polynomial proved nonnegative by exact rational affine squares plus nonnegative monomials. The independent checker expands these certificates, substitutes them back, checks every degree bound, and independently collects the complete quartic. The literal TeX residuals are parsed back and compared term by term.

`verify.py --rebuild` checks all endpoints without mutating them and rebuilds the selected mathematical files in temporary directories. The portable source, exact source JSON, retained ordinary proofs and named Lean calibration inputs are included. Source-path hashes and modifications are exposed in `SOURCE_MAP.json`; the full unbounded new argument is also available as `PROOF.md`. The supplied calibration inputs do not constitute a newly rebuilt full transitive Lean development or an encoded object-language ZFC proof.

5 Ten-positive equations by denominator engineering

This section supplies six complete ordinary integer polynomials, each in ten strictly positive integer unknowns. Four give useful local operation/degree tradeoffs; two are retained as dominated

controls. All fixed coefficients are the *whole* indexed BFS943 native tuple, supplied literally in `compact/coefficients/`. Its finite reconstruction is `rebuild_coefficients.py`; its largest port has 592,884 bits. These are not coefficient-free formulas, and fixed ports are not unknowns.

Chart	Weight	Binary operations	Exact degree
$Y = q^3 s,\ D = XY$	V^2	77	626
$Y = q^3 s,\ D = XY$	bV	78	464
$Y = q^3 s,\ D = q^4$	V^2	78	622
$Y = q^3 s,\ D = q^4$	bV	79	460
$Y = q^2 s,\ D = q^3 Y$	V^2	79	518
$Y = q^2 s,\ D = q^3 Y$	bV	80	384

The 78/622 and 79/518 points are dominated by 78/464; they are uniform-family controls, not frontier claims. The incoming unscaled 78/630 and 79/468 points used $D = Y^2$. The new 77/626 and 78/464 points respectively improve both metrics of those matched predecessors. The 80/384 point also improves the incoming unshifted 83/436 point in both metrics. Older lower-degree constructions remain relevant. No global optimum or literature-wide priority is claimed.

Attribution and scope. The incoming *Certificate Reductions* argument already removed the artificial scale and recovered a Pell index from a known divisor before recovering a whole denominator. *Semantic Signs* already proved unshifted transport and the exclusion of both signed third auxiliary branches. Pell doubling, product-square extraction, and quotient normalization are inherited. Here the new choices are $D = XY$, $D = q^4$, and the partially normalized $Y = q^2 s, D = q^3 Y$, whose fractional carrier numerator reuses Y^2. The proof below composes explicitly with the retained ordinary Pell/source theorems. It is not a new end-to-end proof-assistant certificate and proves neither consistency nor unsolvability separately.

5.1 The complete polynomials

The only independent coordinates are

$$J, F, \alpha, i, T, w, s, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

The fixed positive integers β, K_0, M_C, M_F are the unchanged indexed tuple, with $\beta \geq 31$ odd and $0 < M_C < \beta < M_F < 2\beta$. Define polynomial abbreviations

$$\begin{gathered}
q = \beta J + 1, \quad h = q - 1, \quad C = q - F - \alpha, \quad X = wq, \\
k = \eta + \zeta, \quad c = kY + \eta, \quad A = Y(X+1) + 1, \\
\Delta = A^2 - 1, \quad H = 4A - 5, \quad v = XY^2, \\
E = 1 + v(v+1)k^2, \\
R = [h(q-F) + \alpha](q^2 - 1) + (M_C + qM_F)J, \\
e = k - R - 1, \quad B = (K_0 + w)C - F, \\
U = ic^2, \quad a = \Delta U^2 E, \quad F_0 = 1 + a, \quad G = EF_0, \\
F_{\mathrm{sq}} = 1 + 4aF_0, \quad Q = \Delta(F_{\mathrm{sq}} - 1), \\
b = cT - R, \quad V = F_{\mathrm{sq}} b - c, \quad K = X(3c - 2X) + 2.
\end{gathered}$$

Choose (Y, D) from $(q^3 s, XY), (q^3 s, q^4), (q^2 s, q^3 Y)$ and choose $L = V^2$ or $L = bV$. Put $M = hDH$. In the first two charts set

$$Z = hD\{H(n - LB) + K\} - H(DB + he).$$

In the partial chart $Y = q^2 s, D = q^3 Y$ set instead

$$Z = hD\{H(n - LB) + K\} + H\{h(Y^2 - e) - DB\}.$$

In every case the complete polynomial equation is

$$P = Q(Z^2 - M^2 G V^2) + (M^2 G - Z^2) = 0. \tag{5}$$

Every operation is addition, subtraction or multiplication; no division, root, exponent variable or hidden quantified intermediate occurs. For proof only let $z = Z/M$, $\kappa = K/H$, $\rho = e/D$, $t = B/h$, $W = 1 + hL$, and let $\sigma = 0$ in the cubic charts and $\sigma = s/q$ in the partial chart. The exact rational identity is

$$z = n + \kappa + \sigma - Wt - \rho. \tag{6}$$

In the partial chart $D\sigma = Y^2$ is the crucial reuse identity.

Theorem 5.1. *Each of the six polynomials above has a strictly positive integer zero if and only if the retained syntactic calculus has a finite ZFC refutation. Consequently, by the retained whole indexed calibration, ZFC proves that its positive unsolvability is equivalent to* Con(ZFC).

Proof. We give the changed unbounded arithmetic argument, retaining the stated universal Pell, ratio and native-calibration interfaces. Write $\psi_A(r), \chi_A(r)$ for Pell ordinate and trace.

Roots before signs. Positive inputs imply $q, X, Y \geq 32$, $c > k \geq 2$, $A > 4$, $U \geq c^2$, $E > 1$, $a > c$, $Q > c^2$, and $M > 0$. The identities

$$F_{\mathrm{sq}} = (1 + 2a)^2, \qquad Q = 4\Delta^2 U^2 G, \qquad \gcd(E, F_0) = 1$$

are unconditional. For any integral b, $V \neq 0$ and $QV^2 - 1 > 0$. Equation (5) gives $(Q - 1)z^2 = G(QV^2 - 1)$. Since $G \mid Q$, both other integer factors are -1 modulo G; prime valuations force $G = g^2$. Coprimality gives $E = \tau^2, F_0 = f_0^2, g = \tau f_0$ with positive roots. The rational number $y = |z|/g$ satisfies

$$QV^2 - (Q - 1)y^2 = 1. \tag{7}$$

Rational adjacent-Pell descent makes y integral: if its reduced denominator is m, then $m^2 \mid Q - 1$; the norm-preserving map

$$(v, y) \longmapsto ((2Q - 1)v - 2(Q - 1)y, (2Q - 1)y - 2Qv)$$

keeps the first coordinate integral, keeps the reduced denominator, and strictly decreases $|v| > 1$ after sign reflection. At $|v| = 1$, $y = 1$. Thus $z = \varepsilon g y$ is integral.

Put $f = 1 + 2a, I = 2Ug$. Then $f^2 = F_{\mathrm{sq}}$, $f^2 - \Delta I^2 = 1$, $c^2 \mid I$ and $Q = (\Delta I)^2$. The retained signed all-index folding theorem, which allows either sign of V, applies to these equations and $V = -c \pmod f$. It gives $c = \psi_A(p)$ for an odd $p \geq 1$. The first norm gives $k = 2\psi_{2v+1}(n_0)$, $n_0 \geq 1$. The value $n_0 = 1$ would give $c = 2Y + 1$ strictly between $\psi_A(1) = 1$ and $\psi_A(2) = 2A$. Hence

$$k \geq 8XY^2 + 4, \qquad c > kY > 8XY^3 \geq 8q^4, \qquad \kappa > 0. \tag{8}$$

No sign of B, C, R, b has been used.

Sign-free transport. At a zero, $b = 0$ would make $y^2 = c^2 + (c^2 - 1)/(Q - 1)$ lie between consecutive integers. Thus $\operatorname{sgn} V = \operatorname{sgn} b$ and $|V| > |b|$. Also $a^2 - G = E[\Delta U^2 E(\Delta U^2 - 1) - 1] > 0$, $F_{\mathrm{sq}} - c > 3a > 3g$, and $y < 3|V|/2$. Therefore $gy < V^2/2$.

For $L = bV$ a stronger bound is needed. Suppose $p = 3$, so $c = 4A^2 - 1$. If $n_0 \geq 3$, then $kY > 32X^2Y^5$, whereas $c < 16X^2Y^2$, a contradiction. Hence $n_0 = 2$ and $k = 8XY^2 + 4$. For the integer $u = X + 2 - 2Y$ one obtains

$$\eta = 4Y^2(Xu + 1) + 8Y(X + 1) + 3 - 4Y.$$

If $u \leq -1$ this is negative, since $Y(X - 1) > 2(X + 1)$ for $X, Y \geq 32$. If $u \geq 2$ it exceeds k. Hence $u \in \{0, 1\}$, or $2Y - X \in \{1, 2\}$, impossible since $q \mid X, Y$. Thus $p \geq 5$ and $c > c_3 = 4\Delta + 3$.

Writing $S = \Delta I$, Pell classification of (7) gives $|V| = \chi_S(j)/S$ for odd j. The first possibility is $|V| = 1$, already excluded. For the third, $V = \pm(4Q - 3)$ gives respectively

$$F_{\mathrm{sq}}(b - 4\Delta) = c - c_3, \qquad F_{\mathrm{sq}}(b + 4\Delta) = c + c_3.$$

Both right sides lie strictly between 0 and F_{sq}. Thus $j \geq 5$, $|V| > 8Q^2$, and, since $Q > F_{\mathrm{sq}} > 2c$,

$$|b| > \frac{8Q^2 - c}{F_{\mathrm{sq}}} > 4F_{\mathrm{sq}} > 4g.$$

Consequently $L = bV = |b||V| > 0$ and $gy < 3L/8 < L/2$.

For either weight $e < b$ because $b - e = cT - k + 1 > 0$, so $\rho < |b|/D$. If $B \leq -1$, then $Wt < -L$; the carrier gives

$$n < L/2 - L + |b|/D < 0,$$

using $D > 2$, $|V| > |b|$ and $|V| > 1$. Contradiction. Thus $B \geq 0$, and $F > 0$ forces $C > 0$. Only now use the mask estimate $3q < R < q^4 - 2$. Equation (8) gives $c > 2R$, hence $b > 0, V > 0$. The retained normalized-rank and signed-index theorems, using $c^2 \mid I$, $c > 2R, c > 2p$ and the fixed-minus residues, give $p = R$. Their hypotheses do not require the removed shift $T + \alpha$. Odd R then gives even q and odd J.

Recover a small index before the full denominator. The recurrence at $P_0 = 2XY^2 + 1$ gives

$$k \equiv 2n_0 \pmod{4XY^2}.$$

Since $P_0 > A$, $n_0 \geq R$ would imply $k/2 \geq c$. Thus $1 \leq n_0 < R$. Integral z gives $D \mid hHe$ in the cubic charts, where $q^4 \mid D$. Since $H = -1 \pmod q$ and $h = q - 1$, this yields $q^4 \mid e$. In the partial chart clearing qhH gives $D \mid qhHe$; $q^5 \mid D$ again gives $q^4 \mid e$. Crucially $q^4 \mid 4XY^2$ in *all* these charts. Therefore $q^4 \mid 2n_0 - R - 1$, and $|2n_0 - R - 1| < R < q^4$ forces $2n_0 = R + 1$. The full recurrence now gives $4XY^2 \mid e$. Each selected D divides $4XY^2$; for $D = q^3Y$ the quotient is $4ws$. Thus ρ is integral. No false coprimality of Y and h was invoked.

Projection and source decoding. Write $R = 2r + 1$, so $r \geq 48$. The retained ratio theorem at scale one, with $\xi = (X + 1)^{2r}/X^r$, gives

$$\xi < \frac{\psi_A(2r + 1)}{\psi_{P_0}(r + 1)} < \xi \left(1 + \frac{1}{Y(X + 1)}\right)^{2r}.$$

Together with $Y < c/k < Y + 1$, this gives $Y > X^r/3$ and $X^{r+1} < H$. With ρ integral the carrier gives $H \mid hK$ in the cubic charts, $H \mid qhK$ in the partial chart. Either implies the inherited premise $H \mid q^3hK$. Universal projection and odd H give

$$H \mid q^3h(X - 2^R)(2^RX + 1).$$

The retained whole-product bound puts the absolute value of this *entire* product below $X^{r+1} < H$, hence $X = 2^R$. In particular κ is integral; no individual nonunit factor was cancelled modulo H.

The cubic carrier gives $h \mid B$. The partial carrier gives $s/q - WB/h \in \mathbb{Z}$; coprimality of q, h gives $q \mid s$ and $h \mid B$. Thus $q^3 \mid Y$ and $t \geq 0$ is integral in every chart. The retained fixed-minus parity and half-binomial rounding theorems now have their full hypotheses. Their fresh reconstruction at $A_{\text{source}} = Y(X+1)+2$, followed by the whole indexed native calibration, supplies a finite ZFC refutation.

Positive converse. A refutation supplies the retained canonical outer data, $X = 2^R$, $R \equiv 3 \pmod 4$, $R > 3q$, $q^3 \mid Y$, integral $t \geq 0$, and the required half-binomial Y. Set $s = Y/q^3$ in cubic charts, $s = Y/q^2$ in the partial chart. Form the scale-one bases and canonical $c = \psi_A(R)$, $k = 2\psi_{P_0}((R+1)/2)$, $\tau = \chi_{P_0}((R+1)/2)$. The ratio/rounding theorem gives $0 < \eta = c - kY < k$, so $\zeta = k - \eta > 0$. Also $4XY^2 \mid e$, hence $\rho \geq 0$ is integral, and projection gives integral $0 < \kappa < Xc$. Choose a positive matrix-period multiple at A modulo $c^2\tau$, with seed pair (f_0, I_0) and $c^2\tau \mid I_0$. Put $i = I_0/(c^2\tau) > 0$ and double the pair; this produces exactly $f = 2f_0^2 - 1$, $I = 2f_0 I_0$, and all literal root definitions above. The retained auxiliary completeness theorem supplies positive V, y and

$$T = \frac{V + c + Rf^2}{cf^2} \in \mathbb{Z}_{>0}.$$

No subtraction of α is needed. Set $n = gy + Wt + \rho - \kappa - \sigma$. In the partial chart $\sigma = Y/q^3$ is now integral; in cubic charts it is zero. Since $\sigma < Y < Xc$, $\kappa < Xc$, and $f_0 > (A-1)I_0 > 2Xc$, we have $g > \kappa + \sigma$ and therefore $n > 0$. The carrier is $z = gy$, proving (5). All independent coordinates are positive; this is a fresh witness construction, not an unjustified positive inverse shear. □

5.2 Exact degree and reproducibility

Let $q_0 = \beta J$, $k_0 = \eta + \zeta$, $C_0 = q_0 - F - \alpha$. The full highest homogeneous forms are respectively

$$\begin{aligned}
(XY, V^2): &\quad 2^{14} q_0^{378} w^{48} s^{110} i^{20} k_0^{64} T^4 C_0^2, \\
(XY, bV): &\quad -2^{10} q_0^{284} w^{36} s^{82} i^{14} k_0^{46} T^2, \\
(q^4, V^2): &\quad 2^{14} q_0^{378} w^{46} s^{108} i^{20} k_0^{64} T^4 C_0^2, \\
(q^4, bV): &\quad -2^{10} q_0^{284} w^{34} s^{80} i^{14} k_0^{46} T^2, \\
(q^3 Y, V^2): &\quad 2^{14} q_0^{272} w^{46} s^{110} i^{20} k_0^{64} T^4 C_0^2, \\
(q^3 Y, bV): &\quad -2^{10} q_0^{206} w^{34} s^{82} i^{14} k_0^{46} T^2.
\end{aligned}$$

For V^2 the unique top summand is QZ^2; for bV it is $-QM^2GV^2$. The displayed nonzero forms prove the claimed exact global degrees at the actual positive β. Every binary operation is charged, including squares and multiplication by scalar literals. All inputs, ports and gates are live. Six deterministic DAGs, frozen parent data, coefficient bytes, recipe and mathematical provenance accompany the proof. The independent checker covers every row by exact polynomial-cut identities, propagates complete leading forms, authenticates all fixed bytes, and rejects 48 deliberate algebraic corruptions. Normal and optimized Python replays and deterministic rebuilds are supplied. These finite checks are not substitutes for the unbounded theorem above.

A tempting $Y = qs, D = q^6$ route at 81/410 or 82/304 was rejected: $4XY^2$ is only visibly divisible by q^3, insufficient for its index argument. It is not an accepted equation or delivered endpoint.

6 Complete fixed-coefficient recipe

The following finite recipe is retained from the inspected ProveIt indexed constant specification, whose exact source and SHA-256 are recorded in `compact/provenance/indexed_constants.tex`

and `compact/SOURCE_PINS.json`. It defines the entire unchanged BFS943 tuple, not a new specialization. The symbol K in the inherited recipe is renamed K_0 to agree with Section 5; symbols in this section are local fixed data, not additional unknowns. The supplied standard-library reconstruction checks every resulting byte against all six literal circuits. The source shift is already included in M_F. **The four fixed integers.** Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1011\} \\ & \cup \{1013, \dots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.\end{aligned}$$

Then

$$\begin{aligned}M_C &= B - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\ M_F &= B - 1 + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) \\ &\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\ &\quad + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805}, \\ K_0 &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ &\quad + (\rho^{4393} + \rho^{5495})\Phi(1/\rho),\end{aligned}$$

where

$$\begin{aligned}\Phi(x) &= (1 + 2^{51}) G_{943}(x) - 2^{10} \sum_{i=0}^{942} v(t_i) x^i \\ &\quad + \sum_{(b,e,m) \in \mathcal{C}} 2^b x^e G_m(2x), \\ \mathcal{C} = {} & \{(10, 1023, 8), (18, 1013, 9), (27, 1003, 8), \\ & (35, 983, 9), (44, 1031, 1), (45, 1022, 1), (46, 1011, 1), \\ & (47, 992, 1), (48, 1095, 1), (49, 2195, 1), (50, 3296, 1)\}.\end{aligned}$$

The rational intermediate in K_0 cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. Let

$$\begin{gathered}h(q, b) = 2(q + 1) + b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\}, \\ \mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\}, \\ r_q = 6 + |\mathcal{R} \cap \{0, \dots, q - 1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0, \dots, q - 1\}|.\end{gathered}$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w, n, e, s)$
$0, 1$	$(2, 435, 3, 1),\ (5, 0, 0, 0)$
$j + 2\ (j = 2q + b < 430)$	$((1 - d_j) l_{u_j}, h(q, b), d_j r_{u_j}, v_j)$
$432 + b\ (b = 0, 1)$	$(0, h(215, b), 0, h(215, b))$
$434 + b\ (b = 0, 1)$	$(0, b, 0, b)$
$424 + 2r_q + b\ (q \in \mathcal{R}, b = 0, 1)$	$(r_q, b, 0, h(q, b))$
$424 + 2l_q + b\ (q \in \mathcal{L}, b = 0, 1)$	$(0, b, l_q, h(q, b))$
$924 + j\ (0 \le j < 4)$	$(0, j + 430\lfloor j/2 \rfloor, 0, 438)$
928	$(2, 434, 2, 0)$
929	$(3, 436, 4, 4)$
930	$(4, 436, 4, 0)$
$497 + t\ (t = 434, 435, 436)$	$(1, 438, 1, t)$
$934+3a+b-6\lfloor a/4 \rfloor$ $a \in \{0,1,4\},\ b=0,1,2$	$(a, 437, b, 437)$

For $t = (w, n, e, s)$ put

$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\, 2^{34}.$$

A closed decoder for the 430 actions. Put $D_b(A, k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430 + q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14(1 - m_q) + m_q D_{16}(E_0, \tau_q),$$
$$u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j) D_{216}(T_0, j - \sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \ldots, 61$, is

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz.
```

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

7 Primitive odd carriers: fourteen integers at degree 86

The auxiliary Pell root and the triangular positivity coordinate can be combined directly, without carrying the first Pell root through a three-root norm. The resulting complete equation has fourteen unrestricted integer unknowns, exact degree 86, and 112 live binary operations. It improves both degree and arithmetic length over the retained 14/88/119 point. A 14/94/110 companion saves two more operations at greater degree. The proof changes no fixed numeral or native/compiler calibration.

Construction	Integer unknowns	Exact degree	Operations
Primitive auxiliary/guard carrier	14	86	112
Primitive strong/guard carrier	14	94	110
Exact recompilation of old two-root equation	14	88	115
Exact recompilation of old three-root equation	13	156	125

The 88/115 row is dominated by 86/112 and is retained to isolate the compiler-only saving. Arithmetic operations mean live binary addition, subtraction and multiplication, including multiplication by constants; these counts do not measure bit complexity.

7.1 Common arithmetic front and the precise inherited interface

The six fixed integers are exactly the retained source34 tuple $(\beta, K_0, M_C, M_F, A_c, B_c)$, with M_F already shifted. In particular $\beta = 2^{1064921} - 1$, $A_c = 2^{1113}$ and $B_c = 2^{666104}$. Their complete authenticated payloads and finite reconstruction recipe accompany the DAGs. Define polynomial abbreviations

$$\begin{gathered}
J = 6j + 1, \quad q_m = \beta J, \quad q = q_m + 1, \\
W = 12w + 4, \quad X = qW, \quad Y = v, \\
F = (K_0 + W)C - zq_m, \\
R = (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J, \\
E = XY, \quad a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \\
k = R + 1 + hE, \quad \eta = c - Yk, \quad L = E(Yk), \\
I = (2r + 1)c^2, \quad M = 2q^3, \quad U = MH, \\
K_\ell = X(2X - 3c) - 2, \quad Z = Un + MK_\ell + H(Y - q^3), \\
D_e = (A_cF - q)^2 + (B_cC - q)^2, \\
\mathcal{B} = (Yk)^2(q^2 - D_e) - q^2(4(Y\eta)^2 + (kq^3)^2), \\
A = (Z + Uk)^2 + 4U(LZ - U).
\end{gathered}$$

All names in this display are polynomial abbreviations, not additional witnesses. The fifteen-integer parent additionally uses f, y, p_1, p_2, p_3 and

$$\begin{aligned}
Q &= \Delta(f^2 - 1), & V &= f^2(cT - R) - c, \\
B_s &= f^2 - \Delta I^2 - 1, & C_a &= V^2 - (Q - 1)y(2V + y) - 1, \\
B_g &= \mathcal{B} - 1 - p_1^2 - p_2^2 - p_3(p_3 + 1).
\end{aligned}$$

Its equation is $A^2 + B_s^2 + C_a^2 + B_g^2 = 0$. The retained ordinary proof identifies integer-zero existence with finite syntactic ZFC refutability. The following new maps restore an actual zero of this precise parent, rather than introducing a new source interpretation.

The facts needed before extraction hold on every integer assignment. The integer q is even and nonzero, H is odd and nonzero, and $U \neq 0$. Also $|X| \geq 8$, so $a = (X+1)Y$ is zero or has $|a| \geq 7$; hence $\Delta = (a+2)^2 - 1 \geq 3$. The polynomial A is a square modulo 4, so $A \neq -2$. Since Δ is 0 or 3 modulo 4, B_s is $0, 1$ or 3 modulo 4, and $B_s \neq -2$ as well.

7.2 An odd carrier with no exceptional zero branch

Lemma 7.1 (Primitive merger)**.** *Let $b, d, e, s \in \mathbb{Z}$, with $d \neq 0$. Set*

$$G = 4b + 1, \qquad K = d\big(s(s+1) - b\big) - e, \qquad N = K^2 - Gde.$$

Then $N = 0$ if and only if there are integers g, u with

$$g^2 = G, \qquad du^2 = e, \qquad 2s + 1 = g + 2u.$$

Proof. Put $x = 2s + 1 \neq 0$. Direct expansion gives

$$16N = \big[d(x^2 + G) - 4e\big]^2 - 4d^2x^2G.$$

At a zero, $g = [d(x^2 + G) - 4e]/(2dx)$ is rational with integral square G, so it is an integer. Since $G \equiv 1 \pmod 4$, g is odd. Thus $u = (x - g)/2$ is integral, and the defining relation for g gives $du^2 = e$. Conversely substitution gives $K = dgu$, hence $N = 0$. The argument permits negative d, e and proves, rather than assumes, nonnegative G. In particular it loses no irrational-cancellation or zero-carrier branch. □

The generic primitive identity is already present in the retained parity-norm literature; the new result is its placement in this free-scale interface and the complete degree/cost reduction.

7.3 The complete degree-86 equation

Use the fourteen unrestricted inputs

$$(j, C, z, h, r, v, w, T, c, n, f, p_2, p_1, s).$$

Keep the front and Q, V above, and define

$$\begin{aligned} &B = \mathcal{B} - p_1^2 - p_2^2, && G = 4B - 3, && \delta = Q - 1, && C_y = QV^2 - 1, \\ &K = \delta\big(s(s+1) - B + 1\big) - C_y, && N = K^2 - G\delta C_y, && S = f^2 - \Delta I^2. \end{aligned}$$

The single equation is

$$\boxed{P_{86} = (A+1)S(N+1) - 1 = 0.} \tag{9}$$

This is an ordinary polynomial with fixed natural powers and no divisions, radicals or hidden witnesses. Its full division-free DAG is supplied.

Proposition 7.2. *Equation (9) has an integer zero if and only if the fixed fifteen-variable parent does.*

Proof. At a zero, the three integer factors $(A+1), S, (N+1)$ are units. The first cannot be -1 because $A \neq -2$, and the second cannot be -1 because $S = B_s + 1$ and $B_s \neq -2$. Thus

$A = B_s = N = 0$. The strong row implies $Q = \Delta^2 I^2$, which is 0 or at least 9; therefore $\delta \neq 0$. Apply the lemma with $(b, d, e) = (B - 1, \delta, C_y)$, obtaining odd g and integral u. Restore

$$p_{3,\mathrm{old}} = (g - 1)/2, \qquad y_{\mathrm{old}} = u - V.$$

All other inputs are unchanged. Then $B_g = 0$ and

$$C_{a,\mathrm{old}} = QV^2 - (Q - 1)(V + y_{\mathrm{old}})^2 - 1 = C_y - \delta u^2 = 0.$$

This includes $I = 0$, where $Q = 0$, $\delta = -1$ and $C_y = -1$.

Conversely, from any parent zero set $s = p_{3,\mathrm{old}} + V + y_{\mathrm{old}}$ and retain all other common inputs. With $g = 2p_{3,\mathrm{old}} + 1$ and $u = V + y_{\mathrm{old}}$, the lemma applies; hence $N = 0$ and the other two factors are 1. These maps work on every zero, not just a canonical family. □

7.4 The degree-94, 110-operation alternative

Use inputs $(j, C, z, h, r, v, w, T, c, n, y, p_1, p_2, s)$ and replace f^2 by

$$D_s = \Delta I^2, \quad F_s = 1 + D_s, \quad Q = \Delta D_s, \quad V = F_s(cT - R) - c.$$

Keep $B = \mathcal{B} - p_1^2 - p_2^2$, $G = 4B - 3$, and put

$$K = s(s + 1) - B - D_s, \quad N = K^2 - GF_s, \quad E_a = V^2 - (Q - 1)y(2V + y).$$

The complete equation is

$$\boxed{P_{94} = (A + 1)(N + 1)E_a - 1 = 0.}$$

Here F_s is 0 or 1 modulo 4, so $N = K^2 - GF_s$ is $0, 1$ or 3 modulo 4 and cannot be -2. The first two factors must consequently both be 1, forcing $A = N = 0$ and $E_a = 1$. Apply the same lemma with $(b, d, e) = (B - 1, 1, F_s)$ to recover $f^2 = F_s$ and odd $g^2 = G$; restore $p_3 = (g - 1)/2$. Conversely set $s = p_3 + f$ at a parent zero. This proves both directions without additional Pell existence assumptions.

7.5 Exact degrees, compiler savings, and verification

Write

$$q_0 = 6\beta j, \quad X_0 = 12q_0 w, \quad Y_0 = v, \quad R_0 = q_0^4 z - 12q_0^3 wC, \quad a_0 = X_0 Y_0,$$
$$U_0 = 8q_0^3 X_0 Y_0, \quad L_0 = X_0 Y_0^2 R_0, \quad Z_0 = U_0 n + 4q_0^3 X_0^2.$$

The complete degree-22 carrier form is $A_{22} = U_0^2 R_0^2 + 4U_0 L_0 Z_0$; the degree-18 budget form is $-R_0^2 q_0^8$. The first carrier summand is the already-disclosed correction to the inherited printed leading form, not a new defect discovery. Its omission does not change the old degree or the following n-dependent coefficient witnesses. The checker propagates the entire actual DAG.

For P_{86} the norm has degree 52, with top form $a_0^4 f^4 R_0^4 q_0^{16}$; the strong unit has degree 12. For P_{94} the norm and auxiliary unit both have degree 36, with top forms $R_0^4 q_0^{16}$ and $32a_0^6 r^4 c^8 R_0 y$. The unique highest product in each finalizer gives

$$[j^{51} w^9 v^{10} z^5 f^4 r^2 c^4 n] P_{86} = -2^{79} 3^{60} \beta^{51} \neq 0,$$
$$[j^{55} w^9 v^{10} z^6 r^4 c^8 y n] P_{94} = 2^{86} 3^{64} \beta^{55} \neq 0.$$

Their exponent sums are 86 and 94, respectively, so the degrees are attained at the actual fixed numeral tuple.

The old 13/156/127 evaluator computed an auxiliary residual $E - 1$ and later added 1 back. Compiling E directly removes two operations, giving 13/156/125 with the identical polynomial. The old 14/88/119 evaluator did the same for both auxiliary and strong residuals; removing both pairs gives 14/88/115, again pointwise-identical.

The independent checker binds every named polynomial cut, all live gates, complete leading forms and actual nonzero coefficients. It checks the primitive and lifting identities, signed and residue controls, complete source/numeral bytes and all 453 separate live-gate corruptions, normally and under Python optimization. A rebuild reproduces every DAG byte for byte. The numeral recipe separately reconstructs all six fixed integers. These are finite algebraic checks, not a decision of zero existence. The unbounded signed-Pell/native/compiler/refutation calibration remains the explicit inherited ordinary dependency; no new end-to-end proof-assistant or object-language consistency certificate is asserted.

8 Reproduction and logical scope

The release includes literal equations, complete fixed coefficient data where used, source provenance, forward and inverse transformations, reconstruction programs, independent exact checks, and the editable sources of this report. The top-level replay command is described in `README.md`. Normal and optimized Python runs check the same mathematical obligations without depending on assertions. Full collected coefficient checks, all-register circuit checks and deliberately corrupted controls supplement the ordinary proofs.

A finite replay cannot certify an unbounded Pell theorem by sampling. The mathematical claims instead compose the displayed universal arguments with the named retained source theorems. The direct-calculus branches use the numerical C222 refutation interface and its ZF–ZFC consistency calibration, not a K215 simulation or Jones universal polynomial. The compact and root-certificate branches retain their distinct native coefficient/refutation calibrations. No tuple from one branch may be substituted into another merely because its port names look similar.

No upstream repository was changed. No literature-wide priority, minimum variable count, lower bound on circuit complexity, or fresh closed-kernel proof is claimed. Source-reviewed improvements arriving after the comparison pin do not retroactively change the identities of the sealed equations or the measured comparisons.

A An alternate coherent radix55 specialization

The separate `compact_radix55/` component retains the six free-port DAGs, domains, counts and degrees, but binds *all four* ports to the accepted current radix55 native instance. The original radix67 component is unchanged. The specialized polynomials differ: their positive β values, and hence their displayed leading coefficients, differ.

This is credited reuse of ProveIt's accepted wrapping-controller encoding, not a new encoding claim. Total raw coefficient storage decreases from 295,388 to 242,484 bytes (52,904 fewer); the largest port decreases from 592,884 to 486,696 bits. No printed-page or PDF-byte gain is claimed. The full independent reconstruction and whole binding are supplied. With the same finite Φ, support and ordered 943 tiles as the fixed recipe, now let $\rho = 2^{55}$, $\mathcal{B} = \rho^{8807}$ and $\beta = \mathcal{B} - 1$. Define

K_0, M_C by the same formal exponent recipes, but replace the *entire* source-mask padding by

$$M_F = \beta + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) + C_{\text{core}} + (\rho - 1) \sum_{j=8792}^{8806} \rho^j + (2^{41} - 4)\rho^{7691}.$$

where $C_{\text{core}} = \sum_{j=1013}^{1022} \rho^j + \sum_{j=1065}^{1073} \rho^j + \rho^{2195} + \rho^{3296}$. Simply changing the radix in the old padding would be wrong.

The bridge uses the complete accepted native theorem, not numerical boxes alone. Its repaired wrapping controller, with the two actual unit residues 3422 and 500242 at 8807 and 581197, constructs fresh canonical words and their *actual* rank with $R \equiv 3 \pmod 4$, $\mathrm{pc}(R) = 3dN + 2$, $q = \mathcal{B}^N$, $X = 2^R$, and Y/q^3 positive odd. Its positive transport coordinate satisfies $B/(q-1) = z_{\text{source}} - 1 \geq 0$. These are exactly the outer data required by the six-chart converse; all seed, strong and auxiliary Pell witnesses are rebuilt afresh. For soundness the six-chart arithmetic restores the same half-binomial, $q^3 \mid Y$, rank and transport fractions. Reconstruct the positive unscaled source at $A_{\text{source}} = Y(X+1) + 2$, choose its raw positive auxiliary quotient T^*. With the fresh source discriminant $\Delta_* = A_{\text{source}}^2 - 1$ and fresh source witnesses i_*, c_*, set $S = \Delta_* i_* c_*^2$. These are not the reduction chart's discriminant or private witnesses. Its accepted P66 factors satisfy $N_0 = N_1 = N_a = N_k = N_t = 1$, $N_s = \Delta_*$, so that whole radix55 polynomial vanishes and its accepted calibration supplies a refutation. No old radix67 native word, isolated coefficient or private witness is reused. The full bridge proof and exact source pins are supplied; the old false modulo 5 unit is not imported into the repaired controller.